NOTES 04: CATEGORICAL VARIABLES

Stat 120 | Spring 2026

Prof Amanda Luby

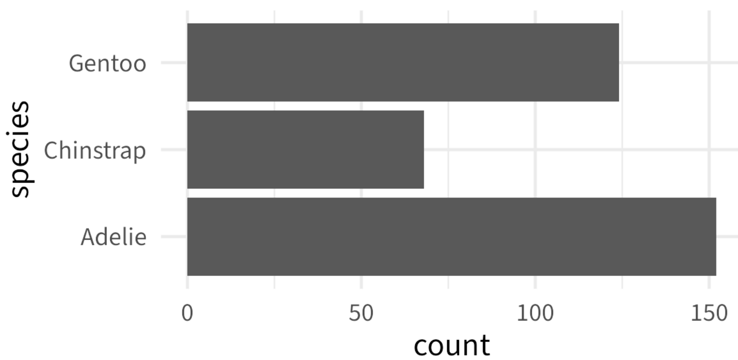
1 Categorical Variables

Categorical variables are best summarized with a **frequency table** and visualized using a **barplot**.

```
table(penguins$species)
```

```
   Adelie Chinstrap    Gentoo 
      152        68       124 
```

```
ggplot(penguins, aes(y = species)) +
  geom_bar()
```



If we want to summarize a categorical variable with a single number, or choose an appropriate parameter/statistic for a categorical variable, we often use a **proportion**.

> Proportion

When we have two categorical variables, we often use a **two-way table** to summarize them at the same time (also called the **joint distribution**).

```
table(penguins$species, penguins$island)
```

```
            Biscoe Dream Torgersen
  Adelie        44    56        52
  Chinstrap      0    68         0
  Gentoo       124     0         0
```

We might also care about the **marginal distribution** (the margins)

```
table(penguins$species, penguins$island) |>
  addmargins()
```

```
            Biscoe Dream Torgersen Sum
  Adelie        44    56        52 152
  Chinstrap      0    68         0  68
  Gentoo       124     0         0 124
  Sum          168   124        52 344
```

or **conditional distribution** (a specific row/column).

```
table(penguins$species, penguins$island) |>
  proportions(2) # condition on COLUMNS
```

```
            Biscoe  Dream Torgersen
  Adelie    0.2619 0.4516    1.0000
  Chinstrap 0.0000 0.5484    0.0000
  Gentoo    0.7381 0.0000    0.0000
```

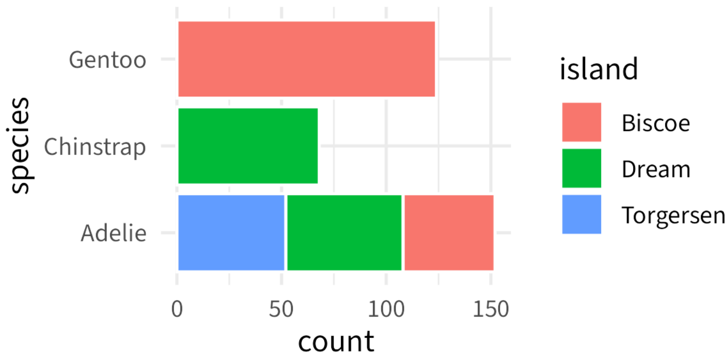
Independence

Two categorical variables are **independent** if the marginal and conditional distributions are the same. This means that knowing something about one variable tells us nothing about the distribution of the other variable.

Example: Is penguin species independent of island?

Visually, we use a filled barplot:

```
ggplot(penguins, aes(y = species, fill = island)) +
  geom_bar(col = "white")
```



Example: Below is the two-way table for our class representing the answers to "What is your class year?" and "Coffee or tea?"

	First year	Sophomore
Coffee	10	8
Neither	3	0
Tea	5	5

a. What is the marginal distribution of *coffee_or_tea*?

b. What is the conditional distribution of *class year* among those who *prefer coffee*?

c. What is the conditional distribution of *coffee or tea* among those who *sophomores*?

d. Does *coffee or tea* appear to be independent of *class year*?